

Laboratorium
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Laporan Akhir

Praktikum Jaringan Komputer

Konfigurasi Vlan, OSPF Multivendor

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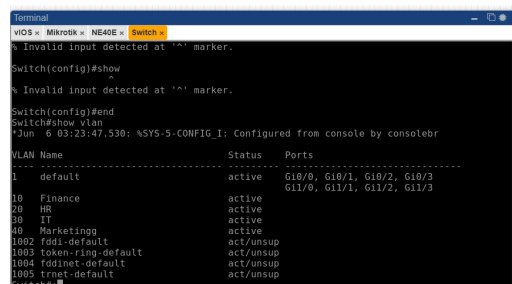
6 Juni 2026

1 Langkah-Langkah Percobaan

Praktikum ini terdiri dari lima bagian, mulai dari konfigurasi switch, router Cisco, router Huawei, MikroTik, sampai pengujian OSPF dan failover.

1.1 Praktikum 1: Konfigurasi Cisco Switch

1. Masuk ke mode konfigurasi switch dengan enable dan configure terminal.
2. Buat VLAN 10 (Finance), VLAN 20 (HR), VLAN 30 (IT), dan VLAN 40 (Marketing), lalu beri nama masing-masing.

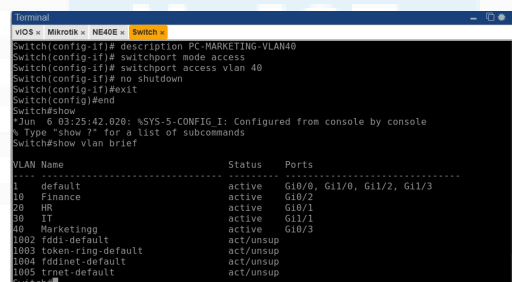


```
Terminal
vIOS > MikroTik > NE40E > Switch
% Invalid input detected at '^' marker.
Switch(config)#show
% Invalid input detected at '^' marker.
Switch(config)#end
Switch#show vlan
*Jun  6 03:23:47.530: %SYS-5-CONFIG_I: Configured from console by consolebr

VLAN Name                Status    Ports
-----
1    default                active    Gi0/0, Gi0/1, Gi0/2, Gi0/3
10   Finance                active    Gi1/0, Gi1/1, Gi1/2, Gi1/3
20   HR                     active
30   IT                     active
40   Marketing              active
1002 fddi-default          act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default        act/unsup
1005 trnet-default          act/unsup
Switch#
```

Gambar 1: Show vlan

3. Cek VLAN yang sudah dibuat dengan show vlan brief, pastikan semuanya aktif.



```
Terminal
vIOS > MikroTik > NE40E > Switch
Switch(config-if)# description PC-MARKETING-VLAN40
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 40
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch#show
*Jun  6 03:25:42.020: %SYS-5-CONFIG_I: Configured from console by console
% Type "show ?" for a list of subcommands
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Gi0/0, Gi1/0, Gi1/2, Gi1/3
10   Finance                active    Gi0/2
20   HR                     active    Gi0/1
30   IT                     active    Gi1/1
40   Marketing              active    Gi0/3
1002 fddi-default          act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default        act/unsup
1005 trnet-default          act/unsup
Switch#
```

Gambar 2: Show vlan Brief

4. Atur port yang mengarah ke PC sebagai access port sesuai VLAN-nya:
 - Gi0/2 → VLAN 10 (PC Finance)
 - Gi0/1 → VLAN 20 (PC HR)
 - Gi1/1 → VLAN 30 (PC IT)
 - Gi0/3 → VLAN 40 (PC Marketing)
5. Atur port Gi0/0 sebagai trunk menuju Cisco Router, dengan encapsulation dot1Q dan membawa VLAN 10, 20, 30, 40.

```

Terminal
vIOS < Mikrotik < NE40E < Switch <
% Invalid input detected at '^' marker.
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#Switch#conf t
% Invalid input detected at '^' marker.
Switch(config)#
Switch(config)#interface gi0/0
Switch(config-if)# switchport trunk encapsulation dot1q
Switch(config-if)# description TRUNK-TO-CISCO-ROUTER
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20,30,40
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#show interfaces trunk
% Invalid input detected at '^' marker.
Switch(config)#exit
Switch# description TRUNK-TO-CISCO-ROUTER

```

Gambar 3: Atur Gi0/0

6. Cek trunk dengan show interfaces trunk.

```

Terminal
vIOS < Mikrotik < NE40E < Switch <
Switch(config-if)# no shutdown
Switch(config-if)#exit
Switch(config)#show interfaces trunk
% Invalid input detected at '^' marker.
Switch(config)#exit
Switch# description TRUNK-TO-CISCO-ROUTER
*Jun  6 03:28:02.766: %SYS-5-CONFIG-I:
Switch#show interfaces trunk

```

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	1
Port	Vlans allowed on trunk			
Gi0/0	10,20,30,40			
Port	Vlans allowed and active in management domain			
Gi0/0	10,20,30,40			
Port	Vlans in spanning tree forwarding state and not pruned			
Gi0/0	10,20,30,40			

Gambar 4: Cek Trunk

7. Simpan konfigurasi dengan write memory.

1.2 Praktikum 2: Konfigurasi Cisco Router

1. Masuk ke mode konfigurasi router.
2. Atur interface Gig0/2 (yang terhubung ke switch) sebagai trunk tanpa IP address.
3. Buat subinterface Gig0/2.10, .20, .30, .40 dengan encapsulation dot1Q sesuai nomor VLAN, lalu kasih IP gateway masing-masing (192.168.X.1/24).
4. Aktifkan DHCP server untuk VLAN 10 dan 20, lengkap dengan pool, default gateway, DNS, dan IP yang dikecualikan.
5. Set IP statis pada PC VLAN 30 dan 40, sedangkan PC VLAN 10 dan 20 diset mode DHCP.
6. Cek interface dengan show ip interface brief, cek pool DHCP dengan show ip dhcp pool, dan cek client yang sudah dapat IP dengan show ip dhcp binding.

```

Terminal
VPC3 < VPC4 < VPC2 < VPC1 < VPC3 <
Router(config)#show ip interface brief
% Invalid input detected at '^' marker.
Router(config)#exit
Router#conf t
*Jun  6 03:36:43.013: %SYS-5-CONFIG-I: Configured from console by c
Router#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	administratively down	down
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	up	up
GigabitEthernet0/2.10	192.168.10.1	YES	manual	up	up
GigabitEthernet0/2.20	192.168.20.1	YES	manual	up	up
GigabitEthernet0/2.30	192.168.30.1	YES	manual	up	up
GigabitEthernet0/2.40	192.168.40.1	YES	manual	up	up
GigabitEthernet0/3	unassigned	YES	unset	administratively down	down

Gambar 5: Show ip Interface brief

```

Terminal
VPC3 < VPC4 < VPC2 < VPC1 < VIOS <
GigabitEthernet0/3      unassigned      YES unset administratively down down
Router#show ip dhcp pool

Pool VLAN10-FINANCE :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 254
  Leased addresses                : 1
  Pending event                  : none
  1 subnet is currently in the pool :
    Current index      IP address range      Leased addresses
    192.168.10.12      192.168.10.1 - 192.168.10.254      1

Pool VLAN20-HR :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 254
  Leased addresses                : 1
  Pending event                  : none
  1 subnet is currently in the pool :
    Current index      IP address range      Leased addresses
    192.168.20.12      192.168.20.1 - 192.168.20.254      1
Router#

```

Gambar 6: Show ip dhcp pool

```

Terminal
VPC3 < VPC4 < VPC2 < VPC1 < VIOS <
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 254
  Leased addresses                : 1
  Pending event                  : none
  1 subnet is currently in the pool :
    Current index      IP address range      Leased addresses
    192.168.10.12      192.168.10.1 - 192.168.10.254      1

Pool VLAN20-HR :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 254
  Leased addresses                : 1
  Pending event                  : none
  1 subnet is currently in the pool :
    Current index      IP address range      Leased addresses
    192.168.20.12      192.168.20.1 - 192.168.20.254      1
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration      Type
                Hardware address/
                User name
192.168.10.11    0100.5070.6668.d6 Jun 07 2026 03:36 AM Automatic
192.168.20.11    0100.5070.6668.d5 Jun 07 2026 03:36 AM Automatic
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration      Type
                Hardware address/
                User name
192.168.10.11    0100.5070.6668.d6 Jun 07 2026 03:38 AM Automatic
192.168.20.11    0100.5070.6668.d5 Jun 07 2026 03:38 AM Automatic
Router#

```

Gambar 7: Show ip dhcp binding

7. Uji dari tiap PC: minta IP via DHCP, ping ke gateway masing-masing, lalu coba ping antar-VLAN.

```

Terminal
VPC3 < VPC4 < VPC2 < VPC1 < VIOS <
VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit: wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team.

Press '?' to get help.

VPCS> ip dhcp
DORA IP 192.168.10.11/24 GW 192.168.10.1

VPCS> ip dhcp
DORA IP 192.168.10.11/24 GW 192.168.10.1

VPCS> ping 192.168.10.1
84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=1.820 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=1.433 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=1.480 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=1.818 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=1.816 ms
VPCS>

```

Gambar 8: IP DHCP VLAN10

```

Terminal
VPC3 < VPC4 < VPC2 < VPC1 < VIOS <
VPCS> ping 192.168.10.1
84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=1.820 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=1.433 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=1.480 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=1.818 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=1.816 ms
VPCS> show ip
NAME      : VPCS[1]
IP/MASK    : 192.168.10.11/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE  : 86342, 86480/43200/75600
MAC        : 00:50:79:66:68:d6
LPORT      : 28000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500
VPCS>

```

Gambar 9: Show ip

```

Terminal
VPC3 < VPC4 < VPC2 < VPC1 < vIOS <

Welcome to Virtual PC Simulator, version 1.0 (0.8c)
Dedicated to Daling.
Build time: Dec 31 2016 01:22:17
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS> ip dhcp
ODORA IP 192.168.20.11/24 GW 192.168.20.1

VPCS>

```

Gambar 10: IP DHCP VLAN20

```

Terminal
VPC3 < VPC4 < VPC2 < VPC1 < vIOS <

VPCS> show ip

NAME      : VPCS[1]
IP/MASK    : 192.168.20.11/24
GATEWAY    : 192.168.20.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE  : 86380, 86400/43200/75600
MAC        : 88:59:79:66:68:d5
LPORT      : 28000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=1.622 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=1.321 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=1.308 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=2.046 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=1.726 ms

VPCS>

```

Gambar 11: Show ip dan ping 192.168.20.1

```

Terminal
VPC3 < VPC4 < VPC2 < VPC1 < vIOS <

ip [all]      Show IPv4 details
              Shows VPC Name, IP address, mask, gateway, DNS, MAC,
              lport, rhost:port and MTU
ipv6 [all]    Show IPv6 details
              Shows VPC Name, IPv6 addresses/mask, gateway, MAC,
              lport, rhost:port and MTU
version       Show the version information

Notes:
1. If no parameter is given, the key information of the current VPC will be
   displayed
2. If 'all' parameter is given for ip/ipv6 a reduced view in tabular
   format will be displayed.

VPCS> ping 192.168.30.1

84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=1.639 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=1.532 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=1.752 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=2.127 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=2.028 ms

VPCS>

```

Gambar 12: ping 192.168.30.1

```

Terminal
VPC3 < VPC4 < VPC2 < VPC1 < vIOS < NE40E < Mikrotik < VPC2 <

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Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS> 192.168.40.10/24 192.168.40.1
Bad command: "192.168.40.10/24 192.168.40.1". Use ? for help.

VPCS> set ip 192.168.40.10/24 192.168.40.1
Invalid command.

VPCS> ip 192.168.40.10/24 192.168.40.1
Checking for duplicate address...
PC1 : 192.168.40.10 255.255.255.0 gateway 192.168.40.1

VPCS> ping 192.168.40.1

84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=1.683 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=1.477 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=1.736 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=1.502 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=1.011 ms

VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=3.428 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=3.111 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.220 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.266 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=5.135 ms

VPCS>

```

Gambar 13: ping 192.168.40.1

```
Terminal
VPC3 < VPC4 < VPC2 < VPC1 < VIOS < NE40E < Mikrotik < VPC0 <

VPC3> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=2.162 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=2.778 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=2.033 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=2.192 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=2.105 ms

VPC3> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=3.024 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=3.197 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=1.784 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=2.249 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=2.572 ms

VPC3> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=3.109 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=2.709 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=3.173 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=2.421 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=2.640 ms

VPC3> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=0.922 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=4.636 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.330 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=7.060 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=5.090 ms

VPC3>
```

Gambar 14: ping 192.168.20.11, 192.168.30.10, 192.168.40.10

```
Terminal
VPC3 < VPC4 < VPC2 < VPC1 < VIOS < NE40E < Mikrotik < VPC0 <

VPC3> show ip dhcp binding
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
192.168.10.12 192.168.10.1 - 192.168.10.254 1

Pool VLAN99-HR :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
192.168.20.12 192.168.20.1 - 192.168.20.254 1

Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address Client-ID/ Hardware address/ Lease expiration Type
192.168.10.11 0100.5070.6668.d6 Jun 07 2026 03:36 AM Automatic
192.168.20.11 0100.5070.6668.d5 Jun 07 2026 03:36 AM Automatic
Router#
```

Gambar 15: show ip dhcp binding

1.3 Praktikum 3: Konfigurasi Huawei Router dan IP Antarrouter

1. Set IP LAN Huawei (Ethernet1/0/2) jadi 192.168.50.1/24.

```
Terminal
VPC3 < VPC4 < VPC2 < VPC1 < VIOS < NE40E < Mikrotik < VPC0 <

<HUAWEI>system-view
Enter system view, return user view with return command.
<HUAWEI>
<HUAWEI>interface Ethernet1/0/2
<HUAWEI-Ethernet1/0/2> description LAN-HUAWEI
<HUAWEI-Ethernet1/0/2> ip address 192.168.50.1 255.255.255.0
<HUAWEI-Ethernet1/0/2> undo shutdown
<HUAWEI-Ethernet1/0/2> quit
<HUAWEI>commit
<HUAWEI>system-view
Error: Unrecognized command found at '^' position.
<HUAWEI>
<HUAWEI>interface Ethernet1/0/0
<HUAWEI-Ethernet1/0/0> description LINK1-T0-CISCO
<HUAWEI-Ethernet1/0/0> ip address 10.10.10.2 255.255.255.252
<HUAWEI-Ethernet1/0/0> undo shutdown
<HUAWEI-Ethernet1/0/0> quit
<HUAWEI>commit
<HUAWEI>
<HUAWEI>
```

Gambar 16: Set IP LAN Huawei

2. Set IP statis PC LAN Huawei (192.168.50.10/24), lalu coba ping ke gateway untuk memastikan sudah konek.
3. Set IP link Cisco–Huawei link 1 (10.10.10.0/30) di Gi0/3 (Cisco) dan Ethernet1/0/0 (Huawei).

```

VPC3 < VPC4 < VPC2 < VPC1 < vIOS < NE40E < Mikrotik < VPC0 <
~*HUAWEI~system-view
Enter system view, return user view with return command.
~*HUAWEI~
~*HUAWEI~interface Ethernet1/0/0
~*HUAWEI~Ethernet1/0/0 description LINK1-TO-CISCO
~*HUAWEI~Ethernet1/0/0 ip address 10.10.2 255.255.252
~*HUAWEI~Ethernet1/0/0 undo shutdown
~*HUAWEI~Ethernet1/0/0 quit
~*HUAWEI~
~*HUAWEI~commit
~*HUAWEI~
~*HUAWEI~system-view
~*HUAWEI~
Error: Unrecognized command found at '^' position.
~*HUAWEI~
~*HUAWEI~interface Ethernet1/0/0
~*HUAWEI~Ethernet1/0/0 description LINK1-TO-CISCO
~*HUAWEI~Ethernet1/0/0 ip address 10.10.2 255.255.252
~*HUAWEI~Ethernet1/0/0 undo shutdown
~*HUAWEI~Ethernet1/0/0 quit
~*HUAWEI~
~*HUAWEI~commit
~*HUAWEI~
~*HUAWEI~

```

Gambar 17: set IP link Cisco 1/0/0

- Set IP link Cisco–Huawei link 2 (10.10.11.0/30) di Gi0/0 (Cisco) dan Ethernet1/0/4 (Huawei).

```

VPC3 < VPC4 < VPC2 < VPC1 < vIOS < NE40E < Mikrotik < VPC0 <
~*HUAWEI~interface Ethernet1/0/0
~*HUAWEI~Ethernet1/0/0 description LINK1-TO-CISCO
~*HUAWEI~Ethernet1/0/0 ip address 10.10.2 255.255.252
~*HUAWEI~Ethernet1/0/0 undo shutdown
~*HUAWEI~Ethernet1/0/0 quit
~*HUAWEI~
~*HUAWEI~commit
~*HUAWEI~
~*HUAWEI~system-view
~*HUAWEI~
Error: Unrecognized command found at '^' position.
~*HUAWEI~
~*HUAWEI~interface Ethernet1/0/0
~*HUAWEI~Ethernet1/0/0 description LINK1-TO-CISCO
~*HUAWEI~Ethernet1/0/0 ip address 10.10.2 255.255.252
~*HUAWEI~Ethernet1/0/0 undo shutdown
~*HUAWEI~Ethernet1/0/0 quit
~*HUAWEI~
~*HUAWEI~commit
~*HUAWEI~
~*HUAWEI~system-view
~*HUAWEI~
Error: Unrecognized command found at '^' position.
~*HUAWEI~
~*HUAWEI~interface Ethernet1/0/4
~*HUAWEI~Ethernet1/0/4 description LINK2-TO-CISCO
~*HUAWEI~Ethernet1/0/4 ip address 10.10.11 255.255.252
~*HUAWEI~Ethernet1/0/4 undo shutdown
~*HUAWEI~Ethernet1/0/4 quit
~*HUAWEI~
~*HUAWEI~commit
~*HUAWEI~
~*HUAWEI~

```

Gambar 18: Set IP Cisco 1/0/4

- Set IP link Huawei–MikroTik (10.10.20.0/30) di Ethernet1/0/3 (Huawei) dan ether1 (MikroTik).
- Set IP link Cisco–MikroTik (10.10.30.0/30) di Gi0/1 (Cisco) dan ether2 (MikroTik).
- Cek semua interface di Cisco, Huawei, dan MikroTik, pastikan statusnya up dan IP-nya sudah benar.

```

VPC3 < VPC4 < VPC2 < VPC1 < vIOS < NE40E < Mikrotik < VPC0 <
down: standby
(l): loopback
(s): spoofing
(d): Dispening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7

Interface          IP Address/Mask    Physical  Protocol VPN
Ethernet1/0/0       10.10.10.2/30      up        up      --
Ethernet1/0/0.4094  128.1.138.137/16   up        up      13vpn
Ethernet1/0/1       128.1.138.137/16   up        down    --
Ethernet1/0/1.4094  128.1.138.137/16   up        up      13vpn
Ethernet1/0/2       192.168.50.1/24    up        up      --
Ethernet1/0/2.4094  128.1.138.137/16   up        up      13vpn
Ethernet1/0/3       10.10.20.1/30      up        up      --
Ethernet1/0/3.4094  128.1.138.137/16   up        up      13vpn
Ethernet1/0/4       10.10.11.2/30      up        up      --
Ethernet1/0/4.4094  128.1.138.137/16   up        up      13vpn
Ethernet1/0/5       unassigned          up        down    --
Ethernet1/0/5.4094  128.1.138.137/16   up        up      13vpn
Ethernet1/0/6       unassigned          up        down    --
Ethernet1/0/6.4094  128.1.138.137/16   up        up      13vpn
Ethernet1/0/7       unassigned          up        down    --
Ethernet1/0/7.4094  128.1.138.137/16   up        up      13vpn
Ethernet1/0/8       unassigned          up        down    --
Ethernet1/0/8.4094  128.1.138.137/16   up        up      13vpn
Ethernet1/0/9       unassigned          up        down    --
Ethernet1/0/9.4094  128.1.138.137/16   up        up      13vpn
GigabitEthernet0/0  unassigned          up        down    --
... More ...

```

Gambar 19: Interface

- Coba ping antar router-router yang bersebelahan untuk pastikan semua link sudah konek sebelum lanjut ke OSPF.

1.3.1 Huawei

```
Terminal
VPC3 < VPC4 < VPC2 < VPC1 < vIOS < NE40E < Mikrotik < VPC0 <
down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7

Interface      IP Address/Mask      Physical      Protocol      VPN
Ethernet0/0/0  10.10.10.2/30         up            up            --
Ethernet0/0/4094 128.1.138.137/16      up            up            13vpn
Ethernet0/0/1  unassigned            up            down          --
Ethernet0/0/1.4094 128.1.138.137/16      up            up            13vpn
Ethernet0/0/2  192.168.50.1/24       up            up            --
Ethernet0/0/2.4094 128.1.138.137/16      up            up            13vpn
Ethernet0/0/3  10.10.20.1/30          up            up            --
Ethernet0/0/3.4094 128.1.138.137/16      up            up            13vpn
Ethernet0/0/4  10.10.11.2/30          up            up            --
Ethernet0/0/4.4094 128.1.138.137/16      up            up            13vpn
Ethernet0/0/5  unassigned            up            down          --
Ethernet0/0/5.4094 128.1.138.137/16      up            up            13vpn
Ethernet0/0/6  unassigned            up            down          --
Ethernet0/0/6.4094 128.1.138.137/16      up            up            13vpn
Ethernet0/0/7  unassigned            up            up            --
Ethernet0/0/7.4094 128.1.138.137/16      up            up            13vpn
Ethernet0/0/8  unassigned            up            down          --
Ethernet0/0/8.4094 128.1.138.137/16      up            up            13vpn
Ethernet0/0/9  unassigned            up            down          --
Ethernet0/0/9.4094 128.1.138.137/16      up            up            13vpn
GigabitEthernet0/0/0 unassigned            up            down          --
--- More ---
```

Gambar 20: Interface

```
Terminal
VPC3 < VPC4 < VPC2 < VPC1 < vIOS < NE40E < Mikrotik < VPC0 <
--HUAWEIping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL C to break
Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=3 ms
Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=1 ms
Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=1 ms
Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=1 ms
Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=2 ms
--- 10.10.10.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/1/3 ms

--HUAWEIping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL C to break
Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=2 ms
Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=1 ms
--- 10.10.11.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/1/2 ms

--HUAWEIping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL C to break
Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=1 ms
--- 10.10.20.2 ping statistics ---
```

Gambar 21: Ping

1.3.2 Cisco

```
Terminal
VPC3 < VPC4 < VPC2 < VPC1 < vIOS < NE40E < Mikrotik < VPC0 <
*Jun 6 03:44:38.631: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Jun 6 03:44:39.632: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config)#interface g10/1
Router(config-if)#description LINK-TO-MIKROTIK
Router(config-if)#ip address 10.10.30.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
*Jun 6 03:46:01.649: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to up
*Jun 6 03:46:02.649: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router(config)#
Router(config)#exit
Router#show
*Jun 6 03:46:29.797: %SYS-5-CONFIG I: Configured from console by console
A Type "show ?" for a list of subcommands
Router#show ip interface brief
% Invalid input detected at '^' marker.

Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  10.10.11.1      YES manual up          up
GigabitEthernet0/1  10.10.30.1      YES manual up          up
GigabitEthernet0/2  unassigned      YES unset  up          up
GigabitEthernet0/2.10  192.168.10.1   YES manual up          up
GigabitEthernet0/2.20  192.168.20.1   YES manual up          up
GigabitEthernet0/2.30  192.168.30.1   YES manual up          up
GigabitEthernet0/2.40  192.168.40.1   YES manual up          up
GigabitEthernet0/3  10.10.10.1      YES manual up          up
Router#display ip interface brief
```

Gambar 22: Interface


```

Terminal
VPC3 = VPC4 = VPC2 = VPC1 = VIOS = NE40E = Mikrotik = VPC0 =

Router#display ip interface brief
% Invalid input detected at '^' marker.
Router#display interface brief
% Invalid input detected at '^' marker.
Router#display ip interface brief
% Invalid input detected at '^' marker.
Router#display ip interface brief
% Invalid input detected at '^' marker.
Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/4 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#

```

Gambar 23: Ping

1.3.3 Mikrotik

```

Terminal
VPC3 = VPC4 = VPC2 = VPC1 = VIOS = NE40E = Mikrotik = VPC0 =

[admin@MikroTik] >> /ip address add address=10.10.20.2/30 interface=ether1 comment=T0-HUAWEI
[admin@MikroTik] >> /ip address add address=10.10.30.2/30 interface=ether2 comment=T0-TISCO
[admin@MikroTik] >> /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 :::: TO-HUAWEI
1 10.10.20.2/30 10.10.20.0 ether1
2 10.10.30.2/30 10.10.30.0 ether2
[admin@MikroTik] >>

```

Gambar 24: Interface

```

Terminal
VPC3 = VPC4 = VPC2 = VPC1 = VIOS = NE40E = Mikrotik = VPC0 =

sent=20 received=20 packet-loss=0% min-rtt=0ms avg-rtt=2ms max-rtt=7ms
SEQ HOST SIZE TTL TIME STATUS
20 10.10.20.1 56 255 1ms
21 10.10.20.1 56 255 1ms
22 10.10.20.1 56 255 1ms
23 10.10.20.1 56 255 2ms
24 10.10.20.1 56 255 1ms
25 10.10.20.1 56 255 1ms
26 10.10.20.1 56 255 1ms
27 10.10.20.1 56 255 1ms
28 10.10.20.1 56 255 1ms
29 10.10.20.1 56 255 1ms
30 10.10.20.1 56 255 0ms
31 10.10.20.1 56 255 1ms
32 10.10.20.1 56 255 1ms
sent=33 received=33 packet-loss=0% min-rtt=0ms avg-rtt=1ms max-rtt=7ms
[admin@MikroTik] > ping 10.10.20.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.20.1 56 255 2ms
1 10.10.20.1 56 255 2ms
2 10.10.20.1 56 255 1ms
3 10.10.20.1 56 255 2ms
sent=4 received=4 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=2ms
[admin@MikroTik] > ping 10.10.30.1
SEQ HOST SIZE TTL TIME STATUS
0 10.10.30.1 56 255 0ms
1 10.10.30.1 56 255 2ms
2 10.10.30.1 56 255 2ms
3 10.10.30.1 56 255 5ms
sent=4 received=4 packet-loss=0% min-rtt=0ms avg-rtt=2ms max-rtt=5ms
[admin@MikroTik] >

```

Gambar 25: Ping

1.4 Praktikum 4: Konfigurasi OSPF Multivendor dan OSPF Cost

1. Konfigurasi OSPF di Cisco Router dengan router-id 1.1.1.1, lalu iklankan semua network link antarrouter dan network VLAN 10, 20, 30, 40.
2. Atur cost OSPF di Cisco: cost 10 untuk dua link ke Huawei (jalur utama), cost 100 untuk link ke MikroTik (jalur backup).

```

Terminal
VPC3 VPC4 VPC2 VPC1 VPC0 NE40E Mikrotik VPC0
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#exit
Router(config)#router ospf 1
Router(config-router)# router-id 1.1.1.1
Router(config-router)# network 10.10.10.0 0.0.0.3 area 0
Router(config-router)# network 10.10.11.0 0.0.0.3 area 0
Router(config-router)# network 10.10.30.0 0.0.0.3 area 0
Router(config-router)# network 192.168.10.0 0.0.0.255 area 0
Router(config-router)# network 192.168.20.0 0.0.0.255 area 0
Router(config-router)# network 192.168.30.0 0.0.0.255 area 0
Router(config-router)# network 192.168.40.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#interface gi0/3
Router(config-if)# description LINK1-TO-HUAWEI
Router(config-if)# ip ospf cost 10
Router(config-if)#exit
Router(config)#
Router(config)#interface gi0/0
Router(config-if)# description LINK2-TO-HUAWEI
Router(config-if)# ip ospf cost 10
Router(config-if)#exit
Router(config)#
Router(config)#interface gi0/1
Router(config-if)# description LINK-TO-MIKROTIK
Router(config-if)# ip ospf cost 100
Router(config-if)#exit
Router(config)#

```

Gambar 26: Konfigurasi OSPF dan atur cost

```

Terminal
VPC3 VPC4 VPC2 VPC1 VPC0 NE40E Mikrotik VPC0
Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 110)

Router#
*Jun  6 03:55:17.025: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from
LOADING to FULL, Loading Done
*Jun  6 03:55:17.025: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 fro
LOADING to FULL, Loading Done
*Jun  6 03:59:00.469: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from
LOADING to FULL, Loading Done
Router#show ip ospf neighbor

Neighbor ID    Pri  State           Dead Time   Address         Interface
3.3.3.3        1    FULL/BDR        00:00:34    10.10.30.2     GigabitEthernet0/1
2.2.2.2        1    FULL/BDR        00:00:35    10.10.11.2     GigabitEthernet0/0
2.2.2.2        1    FULL/BDR        00:00:36    10.10.10.2     GigabitEthernet0/3
Router#

```

Gambar 27: Show ip ospf neighbor

```

Terminal
VPC3 VPC4 VPC2 VPC1 VPC0 NE40E Mikrotik VPC0
Router#show ip ospf neighbor

Neighbor ID    Pri  State           Dead Time   Address         Interface
3.3.3.3        1    FULL/BDR        00:00:38    10.10.30.2     GigabitEthernet0/1
2.2.2.2        1    FULL/BDR        00:00:38    10.10.11.2     GigabitEthernet0/0
2.2.2.2        1    FULL/BDR        00:00:37    10.10.10.2     GigabitEthernet0/3
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:09:25 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:09:25 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:09:35 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, Ex - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NARP, L - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O        10.10.20.0/30 [110/110] via 10.10.11.2, 00:09:35, GigabitEthernet0/0
O        [110/110] via 10.10.10.2, 00:09:45, GigabitEthernet0/3
O        192.168.50.0/24 [110/11] via 10.10.11.2, 00:09:35, GigabitEthernet0/0
O        [110/11] via 10.10.10.2, 00:09:45, GigabitEthernet0/3
Router#di

```

Gambar 28: Show ip route ospf

```

Terminal
VPC3 < VPC4 < VPC2 < VPC1 < VPC0 < NE40E < Mikrotik < VPC0 <
Router#show ip ospf neighbor
Router#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "ospf"
  Sending updates every 0 seconds
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 32
  Routing for Networks:
    Routing Information Sources:
      Gateway         Distance         Last Update
    Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance         Last Update
  Distance: (default is 110)
Router#

```

Gambar 29: Show ip protocols

3. Konfigurasi OSPF di Huawei dengan router-id 2.2.2.2, iklankan link ke Cisco, link ke MikroTik, dan LAN Huawei.
4. Atur cost OSPF di Huawei: cost 10 untuk dua link ke Cisco, cost 100 untuk link ke MikroTik.

```

Terminal
VPC3 < VPC4 < VPC2 < VPC1 < VPC0 < NE40E < Mikrotik < VPC0 <
[~HUAWEI]system-view
Error: Unrecognized command found at '^' position.
[~HUAWEI]system-view
Error: Unrecognized command found at '^' position.
[~HUAWEI]system-view
Error: Unrecognized command found at '^' position.
[~HUAWEI]ospf 1 router-id 2.2.2.2
[~HUAWEI-ospf-1] area 0.0.0.0
[~HUAWEI-ospf-1-area-0.0.0.0] network 10.10.10.0 0.0.0.3
[~HUAWEI-ospf-1-area-0.0.0.0] network 10.10.11.0 0.0.0.3
[~HUAWEI-ospf-1-area-0.0.0.0] network 10.10.20.0 0.0.0.3
[~HUAWEI-ospf-1-area-0.0.0.0] network 192.168.50.0 0.0.0.255
[~HUAWEI-ospf-1-area-0.0.0.0] quit
[~HUAWEI-ospf-1] quit
[~HUAWEI]
[~HUAWEI]interface Ethernet1/0/0
[~HUAWEI-Ethernet1/0/0] description LINK1-T0-CISCO
[~HUAWEI-Ethernet1/0/0] ospf cost 10
[~HUAWEI-Ethernet1/0/0] quit
[~HUAWEI]
[~HUAWEI]interface Ethernet1/0/4
[~HUAWEI-Ethernet1/0/4] description LINK2-T0-CISCO
[~HUAWEI-Ethernet1/0/4] ospf cost 10
[~HUAWEI-Ethernet1/0/4] quit
[~HUAWEI]
[~HUAWEI]interface Ethernet1/0/3
[~HUAWEI-Ethernet1/0/3] description LINK-T0-MIKROTIK
[~HUAWEI-Ethernet1/0/3] ospf cost 100
[~HUAWEI-Ethernet1/0/3] quit
[~HUAWEI]
[~HUAWEI]commit
[~HUAWEI]
[~HUAWEI]

```

Gambar 30: Konfigurasi Ospf dan atur cost Huawei

```

Terminal
VPC3 < VPC4 < VPC2 < VPC1 < VPC0 < NE40E < Mikrotik < VPC0 <
[~HUAWEI-Ethernet1/0/3] description LINK-T0-MIKROTIK
[~HUAWEI-Ethernet1/0/3] ospf cost 100
[~HUAWEI-Ethernet1/0/3] quit
[~HUAWEI]
[~HUAWEI]commit
[~HUAWEI]display ospf peer
(M) Indicates MAD) neighbor

OSPF Process 1 with Router ID 2.2.2.2
Neighbors
Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.10.1
State: Full           Mode: Nbr is Slave   Priority: 1
DR: 10.10.10.1        BDR: 10.10.10.2    MTU: 1500
Dead timer due in 37 sec
Retrans timer interval: 5
Neighbor is up for 00h00m13s
Neighbor Up Time : 2026-06-06 04:19:43
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.11.1
State: Full           Mode: Nbr is Slave   Priority: 1
DR: 10.10.11.1        BDR: 10.10.11.2    MTU: 1500
Dead timer due in 37 sec
Retrans timer interval: 5
Neighbor is up for 00h00m13s
Neighbor Up Time : 2026-06-06 04:19:43
Authentication Sequence: [ 0 ]
[~HUAWEI]

```

Gambar 31: display ospf peer

```

Terminal
VPC3 < VPC4 < VPC2 < VPC1 < vIOS < NE40E < Mikrotik < VPC0 <
Authentication Sequence: [ 0 ]
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public_Routing Table : OSPF
Destinations : 9      Routes : 14

OSPF routing table status : <Active>
Destinations : 5      Routes : 10

Destination/Mask  Proto  Pre  Cost    Flags NextHop        Interface
-----
10.10.30.0/30     OSPF   10   110     D   10.10.11.1   Ethernet1/0/4
192.168.10.0/24   OSPF   10   11      D   10.10.11.1   Ethernet1/0/4
192.168.20.0/24   OSPF   10   11      D   10.10.11.1   Ethernet1/0/4
192.168.30.0/24   OSPF   10   11      D   10.10.11.1   Ethernet1/0/4
192.168.40.0/24   OSPF   10   11      D   10.10.11.1   Ethernet1/0/4
192.168.50.0/24   OSPF   10   11      D   10.10.11.1   Ethernet1/0/4

OSPF routing table status : <Inactive>
Destinations : 4      Routes : 4

Destination/Mask  Proto  Pre  Cost    Flags NextHop        Interface
-----
10.10.10.0/30     OSPF   10   10      T   10.10.10.2   Ethernet1/0/0
10.10.11.0/30     OSPF   10   10      T   10.10.11.2   Ethernet1/0/4
10.10.20.0/30     OSPF   10   100     T   10.10.20.1   Ethernet1/0/3
192.168.50.0/24   OSPF   10   1       T   192.168.50.1 Ethernet1/0/2
[~HUAWEI]

```

Gambar 32: display ip routing-table protocol ospf

```

Terminal
VPC3 < VPC4 < VPC2 < VPC1 < vIOS < NE40E < Mikrotik < VPC0 <
[~HUAWEI]display ospf routing

OSPF Process 1 with Router ID 2.2.2.2
Routing Tables

Routing for Network
Destination      Cost    Type    NextHop    AdvRouter    Area
-----
10.10.10.0/30    10      Direct  10.10.10.2  2.2.2.2      0.0.0.0
10.10.11.0/30    10      Direct  10.10.11.2  2.2.2.2      0.0.0.0
10.10.20.0/30    100     Direct  10.10.20.1  2.2.2.2      0.0.0.0
10.10.30.0/30    110     Stub    10.10.11.1  1.1.1.1      0.0.0.0
10.10.30.0/30    110     Stub    10.10.10.1  1.1.1.1      0.0.0.0
192.168.10.0/24  11      Stub    10.10.11.1  1.1.1.1      0.0.0.0
192.168.10.0/24  11      Stub    10.10.10.1  1.1.1.1      0.0.0.0
192.168.20.0/24  11      Stub    10.10.11.1  1.1.1.1      0.0.0.0
192.168.20.0/24  11      Stub    10.10.10.1  1.1.1.1      0.0.0.0
192.168.30.0/24  11      Stub    10.10.11.1  1.1.1.1      0.0.0.0
192.168.30.0/24  11      Stub    10.10.10.1  1.1.1.1      0.0.0.0
192.168.40.0/24  11      Stub    10.10.11.1  1.1.1.1      0.0.0.0
192.168.40.0/24  11      Stub    10.10.10.1  1.1.1.1      0.0.0.0
192.168.50.0/24  1       Direct  192.168.50.1 2.2.2.2      0.0.0.0

Total Nets: 9
Intra Area: 0 Inter Area: 0 ASE: 0 NSSA: 0

OSPF Process 65534 with Router ID 128.1.138.137
Routing Tables

Routing for Network
Destination      Cost    Type    NextHop    AdvRouter    Area
-----
128.1.138.137/32 0        Direct  128.1.138.137 128.1.138.137 0.0.0.0

Total Nets: 1
Intra Area: 1 Inter Area: 0 ASE: 0 NSSA: 0

```

Gambar 33: display ospf routing

5. Konfigurasi OSPF di MikroTik dengan router-id 3.3.3.3, tambahkan network OSPF untuk link ke Huawei dan ke Cisco, lalu set cost interface jadi 100 sebagai jalur backup.

```

Terminal
VPC3 < VPC4 < VPC2 < VPC1 < vIOS < NE40E < Mikrotik < VPC0 <
syntax error (line 1 column 154)
[admin@MikroTik] >> /routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] >> /routing ospf interface add interface=ether2 cost=100
[admin@MikroTik] >> routing ospf neighbor print
[admin@MikroTik] >> /routing ospf neighbor print
[admin@MikroTik] >> /routing ospf interface add network=10.10.20.0/30 area=backbone
syntax error (line 1 column 37)
[admin@MikroTik] >> /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@MikroTik] >> /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@MikroTik] >> /routing ospf neighbor print
0 instance-default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=0.0.0.0 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=75
1 instance-default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=0.0.0.0 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=35
[admin@MikroTik] >>

```

Gambar 34: Routing ospf neighbor print

```

Terminal
VPC3 < VPC4 < VPC2 < VPC1 < VDS < NE40E < Mikrotik < VPC0 <

1 instance=default router-id=2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=0.0.0.0 state="Full" state-changes=5
ls-retransmit=0 ls-request=0 db-summary=0 adjacency=3s

[admin@Mikrotik] >> /ip route print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, F - rip, B - bgp, O - ospf, M - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 Ado 10.10.10.0/30 10.10.30.1 110
1 Ado 10.10.11.0/30 10.10.20.1 110
2 ADC 10.10.20.0/30 10.10.20.2 ether1 0
3 ADC 10.10.30.0/30 10.10.30.2 ether2 0
4 Ado 192.168.10.0/24 10.10.30.1 110
5 Ado 192.168.20.0/24 10.10.30.1 110
6 Ado 192.168.30.0/24 10.10.30.1 110
7 Ado 192.168.40.0/24 10.10.30.1 110
8 Ado 192.168.50.0/24 10.10.20.1 110

[admin@Mikrotik] >> /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, F - rip, B - bgp, O - ospf, M - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 Ado 10.10.10.0/30 10.10.30.1 110
1 Ado 10.10.11.0/30 10.10.20.1 110
2 Ado 192.168.10.0/24 10.10.30.1 110
3 Ado 192.168.20.0/24 10.10.30.1 110
4 Ado 192.168.30.0/24 10.10.30.1 110
5 Ado 192.168.40.0/24 10.10.30.1 110
6 Ado 192.168.50.0/24 10.10.20.1 110

[admin@Mikrotik] >>

```

Gambar 35: IP route print where ospf

6. Cek neighbor/peer OSPF di tiap router: show ip ospf neighbor (Cisco), display ospf peer (Huawei), /routing ospf neighbor print (MikroTik).

```

Terminal
VPC3 < VPC4 < VPC2 < VPC1 < VDS < NE40E < Mikrotik < VPC0 <

Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 110)

Router#
*Jun  6 03:55:17.025: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from
LOADING to FULL, Loading Done
*Jun  6 03:55:17.025: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 fro
LOADING to FULL, Loading Done
*Jun  6 03:55:19.469: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from
LOADING to FULL, Loading Done
Router#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address         Interface
3.3.3.3        1     FULL/BDR        00:00:34    10.10.30.2      GigabitEthernet0/1
2.2.2.2        1     FULL/BDR        00:00:35    10.10.11.2      GigabitEthernet0/0
2.2.2.2        1     FULL/BDR        00:00:36    10.10.10.2      GigabitEthernet0/3
Router#

```

Gambar 36: show ip ospf neighbor

```

Terminal
VPC3 < VPC4 < VPC2 < VPC1 < VDS < NE40E < Mikrotik < VPC0 <

[*HUAWEI-Ethernet1/0/3] description LINK-TO-MIKROTIK
[*HUAWEI-Ethernet1/0/3] ospf cost 100
[*HUAWEI-Ethernet1/0/3] quit
[*HUAWEI]
[*HUAWEI]commit
[*HUAWEI]display ospf peer

(M) Indicates MADJ neighbor

OSPF Process 1 with Router ID 2.2.2.2
  Neighbors

Area 0.0.0.0 Interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.10.1
State: Full           Mode:Nbr is Slave   Priority: 1
DR: 10.10.10.1        BDR: 10.10.10.2   MTU: 1500
Dead timer due in 37 sec
Retrans timer interval: 5
Neighbor is up for 00h00m13s
Neighbor Up Time : 2026-06-06 04:19:43
Authentication Sequence: [ 0 ]

Area 0.0.0.0 Interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.11.1
State: Full           Mode:Nbr is Slave   Priority: 1
DR: 10.10.11.1        BDR: 10.10.11.2   MTU: 1500
Dead timer due in 37 sec
Retrans timer interval: 5
Neighbor is up for 00h00m13s
Neighbor Up Time : 2026-06-06 04:19:43
Authentication Sequence: [ 0 ]

[*HUAWEI]

```

Gambar 37: display ospf peer

```

Terminal
VPC3 > VPC4 > VPC2 > VPC1 > WDS > NE40E > Mikrotik > VPC0 >
syntax error (line 1 column 154)
[admin@Mikrotik] >> /routing ospf interface add interface=ether1 cost=100
[admin@Mikrotik] >> /routing ospf interface add interface=ether2 cost=100
[admin@Mikrotik] >> /routing ospf neighbor print
[admin@Mikrotik] > /routing ospf neighbor print
[admin@Mikrotik] > /routing ospf interface add network=10.10.20.0/30 area=backbone
syntax error (line 1 column 37)
[admin@Mikrotik] > /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@Mikrotik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@Mikrotik] > /routing ospf neighbor print
0 instance-default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
dr-address=10.10.30.1 backup-dr-address=0.0.0.0 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=75
1 instance-default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=0.0.0.0 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=35
[admin@Mikrotik] >

```

Gambar 38: Routing ospf neighbor print

7. Cek routing table OSPF di tiap router, pastikan semua network sudah saling kebagian (Cisco dapat rute ke LAN Huawei, Huawei dapat rute ke VLAN Cisco, MikroTik dapat rute ke semua network).

```

Terminal
VPC3 > VPC4 > VPC2 > VPC1 > WDS > NE40E > Mikrotik > VPC0 >
10.10.10.0 0.0.0.3 area 0
10.10.11.0 0.0.0.3 area 0
10.10.30.0 0.0.0.3 area 0
192.168.10.0 0.0.0.255 area 0
192.168.20.0 0.0.0.255 area 0
192.168.30.0 0.0.0.255 area 0
192.168.40.0 0.0.0.255 area 0
Routing Information Sources:
Gateway Distance
Distance: (default is 110)
Router#
*Jun 6 03:55:17.025: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from
LOADING to FULL, Loading Done
*Jun 6 03:55:17.025: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 fro
LOADING to FULL, Loading Done
*Jun 6 03:59:09.469: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet0/1 from
LOADING to FULL, Loading Done
Router#show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 1 FULL/BDR 00:00:34 10.10.30.2 GigabitEthernet0/1
2.2.2.2 1 FULL/BDR 00:00:35 10.10.11.2 GigabitEthernet0/0
2.2.2.2 1 FULL/BDR 00:00:36 10.10.10.2 GigabitEthernet0/3
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.11.2 on GigabitEthernet0/0, 00:05:59 ago
Routing Descriptor Blocks:
10.10.11.2, from 2.2.2.2, 00:05:59 ago, via GigabitEthernet0/0
Route metric is 11, traffic share count is 1
* 10.10.10.2, from 2.2.2.2, 00:05:49 ago, via GigabitEthernet0/3
Route metric is 11, traffic share count is 1
Router#

```

Gambar 39: show ip routing

```

Terminal
VPC3 > VPC4 > VPC2 > VPC1 > WDS > NE40E > Mikrotik > VPC0 >
Router#show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 1 FULL/BDR 00:00:38 10.10.30.2 GigabitEthernet0/1
2.2.2.2 1 FULL/BDR 00:00:38 10.10.11.2 GigabitEthernet0/0
2.2.2.2 1 FULL/BDR 00:00:37 10.10.10.2 GigabitEthernet0/3
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.11.2 on GigabitEthernet0/0, 00:09:25 ago
Routing Descriptor Blocks:
10.10.11.2, from 2.2.2.2, 00:09:25 ago, via GigabitEthernet0/0
Route metric is 11, traffic share count is 1
* 10.10.10.2, from 2.2.2.2, 00:09:35 ago, via GigabitEthernet0/3
Route metric is 11, traffic share count is 1
Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR
Gateway of last resort is not set
10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
10.10.20.0/30 [110/110] via 10.10.11.2, 00:09:35, GigabitEthernet0/0
110/110 via 10.10.10.2, 00:09:45, GigabitEthernet0/3
192.168.50.0/24 [110/11] via 10.10.11.2, 00:09:35, GigabitEthernet0/0
110/11 via 10.10.10.2, 00:09:45, GigabitEthernet0/3
Router#di:

```

Gambar 40: Show ip route ospf

```

Terminal
VPC3 > VPC4 > VPC2 > VPC1 > vIOS > NE40E > Mikrotik > VPC0 >
[~HUAWEI]
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public_Routing Table : OSPF
Destinations : 9      Routes : 14

OSPF routing table status : <Active>
Destinations : 5      Routes : 10

Destination/Mask    Proto    Pre    Cost    Flags NextHop        Interface
-----
10.10.30.0/30       OSPF     10     110     D 10.10.11.1    Ethernet1/0/4
192.168.10.0/24     OSPF     10     11      D 10.10.11.1    Ethernet1/0/4
192.168.20.0/24     OSPF     10     11      D 10.10.10.1    Ethernet1/0/0
192.168.30.0/24     OSPF     10     11      D 10.10.11.1    Ethernet1/0/4
192.168.40.0/24     OSPF     10     11      D 10.10.11.1    Ethernet1/0/4
192.168.50.0/24     OSPF     10     11      D 10.10.11.1    Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4      Routes : 4

Destination/Mask    Proto    Pre    Cost    Flags NextHop        Interface
-----
10.10.10.0/30       OSPF     10     10      D 10.10.10.2    Ethernet1/0/0
10.10.11.0/30       OSPF     10     10      D 10.10.11.2    Ethernet1/0/4
10.10.20.0/30       OSPF     10     100     D 10.10.20.1    Ethernet1/0/3
192.168.50.0/24     OSPF     10     1       D 192.168.50.1  Ethernet1/0/2
[~HUAWEI]
[~HUAWEI]

```

Gambar 41: display-ip routing table protocol ospf

```

Terminal
VPC3 > VPC4 > VPC2 > VPC1 > vIOS > NE40E > Mikrotik > VPC0 >
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
dr-address=10.10.20.1 backup-dr-address=0.0.0.0 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=3s
[admin@Mikrotik] >> /ip route print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, F - rip, B - bgp, O - ospf, M - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS      PREF-SRC  GATEWAY      DISTANCE
0 ADO 10.10.10.0/30 10.10.30.1 110
1 ADO 10.10.11.0/30 10.10.30.1 110
2 ADC 10.10.20.0/30 10.10.20.2 0
3 ADC 10.10.30.0/30 10.10.30.2 0
4 ADO 192.168.10.0/24 10.10.30.1 110
5 ADO 192.168.20.0/24 10.10.30.1 110
6 ADO 192.168.30.0/24 10.10.30.1 110
7 ADO 192.168.40.0/24 10.10.30.1 110
8 ADO 192.168.50.0/24 10.10.20.1 110
[admin@Mikrotik] >> /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, F - rip, B - bgp, O - ospf, M - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS      PREF-SRC  GATEWAY      DISTANCE
0 ADO 10.10.10.0/30 10.10.30.1 110
1 ADO 10.10.11.0/30 10.10.30.1 110
2 ADO 192.168.10.0/24 10.10.30.1 110
3 ADO 192.168.20.0/24 10.10.30.1 110
4 ADO 192.168.30.0/24 10.10.30.1 110
5 ADO 192.168.40.0/24 10.10.30.1 110
6 ADO 192.168.50.0/24 10.10.20.1 110
[admin@Mikrotik] >>

```

Gambar 42: IP route print where ospf

8. Coba ping dari tiap PC VLAN ke PC LAN Huawei dan sebaliknya untuk memastikan koneksi end-to-end berhasil.

```

Terminal
VPC3 > VPC4 > VPC2 > VPC1 > vIOS > NE40E > Mikrotik > VPC0 >
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=3.189 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=2.709 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=3.173 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=2.421 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=2.640 ms

VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=9.522 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=4.636 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.330 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=7.060 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=5.090 ms

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 1.935 ms 6.242 ms 3.176 ms
 2 10.10.11.2 5.064 ms 2.918 ms 3.517 ms
 3 *192.168.50.10 3.929 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 1.826 ms 1.723 ms 2.420 ms
 2 10.10.11.2 4.774 ms 2.284 ms 2.157 ms
 3 *192.168.50.10 3.845 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 3.280 ms 3.382 ms 1.671 ms
 2 10.10.11.2 5.250 ms 1.628 ms 2.794 ms
 3 *192.168.50.10 2.439 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>

```

Gambar 43: PC vlan10 Ping 192.168.50.10

```
Terminal
VPC3 < VPC4 < VPC2 < VPC1 < VIOS < NE40E < Mikrotik < VPC0 <

VPCS> ip dhcp
DORA IP 192.168.20.11/24 GW 192.168.20.1

VPCS> show ip

NAME      : VPCS[1]
IP/MASK    : 192.168.20.11/24
GATEWAY    : 192.168.20.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE  : 863809.86480/43280/75600
MAC        : 08:50:79:66:68:d5
I/POR      : 28000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=1.622 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=1.321 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=1.580 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=2.046 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=1.726 ms

VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=3.649 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=3.845 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.953 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=2.745 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=5.345 ms

VPCS>
```

Gambar 44: PC vlan20 Ping 192.168.50.10

```
Terminal
VPC3 < VPC4 < VPC2 < VPC1 < VIOS < NE40E < Mikrotik < VPC0 <

dump      Show dump flags
echo      Show the status of the echo flag. See set echo ?
history   List the command history
ip [all]  Show IPv4 details
          Shows VPC Name, IP address, mask, gateway, DNS, MAC,
          I/por, rhost:rport and MTU
ipv6 [all] Show IPv6 details
          Shows VPC Name, IPv6 addresses/mask, gateway, MAC,
          I/por, rhost:rport and MTU
version   Show the version information

Notes:
1. If no parameter is given, the key information of the current VPC will be
   displayed
2. If 'all' parameter is given for ip/ipv6 a reduced view in tabular
   format will be displayed.

VPCS> ping 192.168.30.1

84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=1.630 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=1.532 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=1.752 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=2.127 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=2.028 ms

VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=3.154 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=7.587 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=12.501 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=6.215 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=3.463 ms

VPCS>
```

Gambar 45: PC vlan30 Ping 192.168.50.10

```
Terminal
VPC3 < VPC4 < VPC2 < VPC1 < VIOS < NE40E < Mikrotik < VPC0 <

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version supporting unetlab by unetlab team

Press '?' to get help.

VPCS> 192.168.40.10/24 192.168.40.1
Bad command: "192.168.40.10/24 192.168.40.1". Use ? for help.

VPCS> set ip 192.168.40.10/24 192.168.40.1
Invalid command.

VPCS> ip 192.168.40.10/24 192.168.40.1
Checking for duplicate address..
PCI : 192.168.40.10 255.255.255.0 gateway 192.168.40.1

VPCS> ping 192.168.40.1

84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=1.683 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=1.477 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=1.736 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=1.502 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=1.611 ms

VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=3.428 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=3.111 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.528 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.266 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=5.135 ms

VPCS>
```

Gambar 46: PC vlan40 Ping 192.168.50.10


```

VPC3 > VPC4 > VPC2 > VPC1 > VIOS > NE40E > Mikrotik > VPC0 >
LPORT : 20000
RHOST:PORT : 127.0.0.1:30000
MTU : 1500

VPCS> ping 192.168.50.1
84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=4.834 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=3.196 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=5.441 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=255 time=2.245 ms
84 bytes from 192.168.50.1 icmp_seq=5 ttl=255 time=2.956 ms

VPCS> ping 192.168.10.11
84 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=4.538 ms
84 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=10.868 ms
^C
VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=62 time=3.748 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=62 time=6.645 ms
^C
VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=6.988 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=15.084 ms
^C
VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=4.799 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=10.750 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=9.908 ms
^C
VPCS>

```

Gambar 47: pengujian dari pc lan huawei ke ip vlan cisco

1.5 Praktikum 5: Pengujian Failover OSPF

1. Kondisi awal (dua link Cisco–Huawei masih hidup): cek rute ke 192.168.50.0/24 dengan `show ip route 192.168.50.0`, lalu trace dari PC VLAN ke LAN Huawei.

```

VPC3 > VPC4 > VPC2 > VPC1 > VIOS > NE40E > Mikrotik > VPC0 >
Router#show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 1 FULL/BDR 00:00:38 10.10.30.2 GigabitEthernet0/1
2.2.2.2 1 FULL/BDR 00:00:38 10.10.11.2 GigabitEthernet0/0
2.2.2.2 1 FULL/BDR 00:00:37 10.10.10.2 GigabitEthernet0/3
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf", distance 110, metric 11, type intra area
Last update from 10.10.11.2 on GigabitEthernet0/0, 00:09:25 ago
Routing Descriptor Blocks:
* 10.10.11.2, from 2.2.2.2, 00:09:25 ago, via GigabitEthernet0/0
Route metric is 11, traffic share count is 1
* 10.10.10.2, from 2.2.2.2, 00:09:35 ago, via GigabitEthernet0/3
Route metric is 11, traffic share count is 1
Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O 10.10.20.0/30 [110/110] via 10.10.11.2, 00:09:35, GigabitEthernet0/0
O 10.10.30.2 [110/110] via 10.10.10.2, 00:09:45, GigabitEthernet0/3
O 192.168.50.0/24 [110/11] via 10.10.11.2, 00:09:35, GigabitEthernet0/0
O 192.168.50.0/24 [110/11] via 10.10.10.2, 00:09:45, GigabitEthernet0/3
Router#dis

```

Gambar 48: show ip route 192.168.50.0

```

NE40E > VPC4 > VPC0 > VPC2 > VPC1 > VIOS >
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.20.1 1.863 ms 1.899 ms 1.659 ms
 2 10.10.10.2 3.725 ms 1.951 ms 1.827 ms
 3 *192.168.50.10 2.342 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.20.1 2.205 ms 1.889 ms 2.103 ms
 2 10.10.11.2 3.546 ms 1.992 ms 2.673 ms
 3 *192.168.50.10 5.013 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.20.1 2.008 ms 2.293 ms 1.523 ms
 2 10.10.30.2 2.889 ms 2.019 ms 1.656 ms
 3 10.10.20.1 4.773 ms 3.144 ms 3.229 ms
 4 *192.168.50.10 6.816 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.20.1 2.323 ms 1.677 ms 1.445 ms
 2 10.10.30.2 2.453 ms 1.928 ms 1.593 ms
 3 10.10.20.1 9.398 ms 2.961 ms 3.764 ms
 4 *192.168.50.10 3.442 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.20.1 2.344 ms 6.521 ms 1.836 ms
 2 10.10.10.2 3.347 ms 2.813 ms 2.926 ms
 3 *192.168.50.10 3.669 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>

```

Gambar 49: trace 192.168.50.0

2. Matikan salah satu link Cisco–Huawei (interface Gi0/3) dengan `shutdown`, lalu cek lagi rute dan trace-nya.

```

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/3
Router(config-if)# shutdown
Router(config-if)#exit
Router(config)#

```

Gambar 50: Matikan 1 link cisco

```

Terminal
VPC3 > VPC4 > VPC2 > VPC1 > VIOS > NE40E > Mikrotik > VPC0 >
10.10.30.0/30 is directly connected, GigabitEthernet0/1
10.10.30.1/32 is directly connected, GigabitEthernet0/1
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
192.168.10.0/24 is directly connected, GigabitEthernet0/2.10
192.168.10.1/32 is directly connected, GigabitEthernet0/2.10
192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
192.168.20.0/24 is directly connected, GigabitEthernet0/2.20
192.168.20.1/32 is directly connected, GigabitEthernet0/2.20
192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
192.168.30.0/24 is directly connected, GigabitEthernet0/2.30
192.168.30.1/32 is directly connected, GigabitEthernet0/2.30
192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
192.168.40.0/24 is directly connected, GigabitEthernet0/2.40
192.168.40.1/32 is directly connected, GigabitEthernet0/2.40
0 192.168.50.0/24 [110/201] via 10.10.30.2, 00:00:26, GigabitEthernet0/1
Router#
Router#
Router#
Router#
Router#
Router#
Router#show ip 192.168.50.0
^
^ Invalid input detected at '^' marker.
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 201, type intra area
Last update from 10.10.30.2 on GigabitEthernet0/1, 00:00:52 ago
Routing Descriptor Blocks:
* 10.10.30.2, from 2.2.2.2, 00:00:52 ago, via GigabitEthernet0/1
Route metric is 201, traffic share count is 1
Router#

```

Gambar 51: show ip route 192.168.50.0

```

Terminal
VPC3 > VPC4 > VPC2 > VPC1 > VIOS > NE40E > Mikrotik > VPC0 >
1 192.168.10.1 2.567 ms 1.879 ms 3.901 ms
2 10.10.30.2 1.961 ms 1.791 ms 1.667 ms
3 10.10.20.1 2.740 ms 2.221 ms 3.891 ms
4 *192.168.50.10 8.478 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
1 192.168.10.1 2.249 ms 2.262 ms 1.721 ms
2 10.10.11.2 3.810 ms 2.682 ms 1.925 ms
3 *192.168.50.10 3.472 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS>

```

Gambar 52: trace 192.168.50.10

- Matikan link Cisco–Huawei yang kedua (Gi0/0), lalu cek apakah rute pindah ke jalur backup lewat MikroTik, dan trace untuk lihat jalur barunya.

```

Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.10.2 on GigabitEthernet0/3, 00:00:06 ago
Routing Descriptor Blocks:
* 10.10.11.2, from 2.2.2.2, 00:00:16 ago, via GigabitEthernet0/0
Route metric is 11, traffic share count is 1
10.10.10.2, from 2.2.2.2, 00:00:06 ago, via GigabitEthernet0/3
Route metric is 11, traffic share count is 1
Router#

```

Gambar 53: show ip route 192.168.50.0

```

Terminal
VPC3 > VPC4 > VPC2 > VPC1 > VIOS > NE40E > Mikrotik > VPC0 >
1 192.168.10.1 2.333 ms 2.101 ms 2.221 ms
2 10.10.30.2 1.978 ms 1.851 ms 2.163 ms
3 *192.168.50.1 3.493 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
1 192.168.10.1 2.567 ms 1.879 ms 3.901 ms
2 10.10.30.2 1.961 ms 1.791 ms 1.667 ms
3 10.10.20.1 2.740 ms 2.221 ms 3.891 ms
4 *192.168.50.10 8.478 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS>

```

Gambar 54: trace 192.168.50.0

- Hidupkan kembali kedua link Cisco–Huawei (no shutdown di Gi0/3 dan Gi0/0), lalu cek apakah rute balik lagi ke jalur utama.

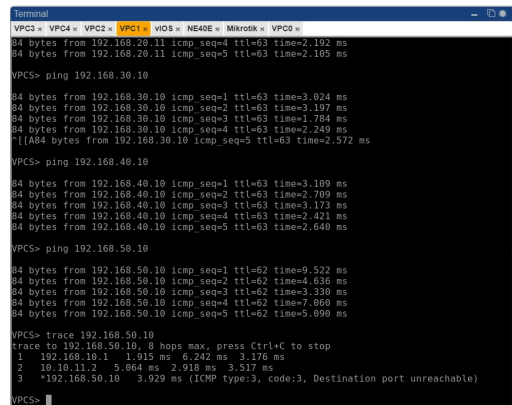
```

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/3
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#interface gi0/0
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#

```

Gambar 55: Dua link cisco-huawei aktifkan lagi

5. Ambil screenshot hasil trace di setiap kondisi: dua link aktif, satu link mati, dan saat traffic lewat MikroTik.



```
Terminal
VPC3 < VPC4 < VPC2 < VPC1 < VIOS < NE40E < Mikrotik < VPC0 <
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=2.102 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=2.105 ms
VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=3.024 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=3.107 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=1.784 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=2.249 ms
*[[84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=2.572 ms
VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=3.189 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=2.709 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=3.173 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=2.421 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=2.640 ms
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=9.522 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=4.636 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.330 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=7.860 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=5.090 ms
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.10.1 1.935 ms 6.242 ms 3.176 ms
 2 10.10.11.2 5.064 ms 2.918 ms 3.517 ms
 3 *192.168.50.10 3.929 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS>
```

Gambar 56: trace 192.168.50.10

2 Analisis Hasil Percobaan

Dari Praktikum 1 dan 2, VLAN 10, 20, 30, dan 40 berhasil dibuat di switch dan diteruskan lewat trunk ke router. Router berhasil menjalankan fungsinya sebagai gateway untuk semua VLAN lewat metode router-on-a-stick (ROAS), di mana satu interface fisik dibagi jadi beberapa subinterface sesuai VLAN. Hal ini terbukti karena semua PC bisa ping ke gateway-nya masing-masing. PC di VLAN 10 dan 20 berhasil dapat IP otomatis dari DHCP server, sedangkan PC VLAN 30 dan 40 pakai IP statis sesuai tabel. Ping antar-VLAN yang berhasil juga membuktikan kalau inter-VLAN routing di router sudah berjalan dengan benar.

Di Praktikum 3, ada perbedaan antara topologi yang digambar dengan kondisi sebenarnya pada link Cisco–Huawei, yaitu Gi0/0 Cisco sebenarnya terhubung ke Ethernet1/0/4 Huawei (link 2), dan Gi0/3 terhubung ke Ethernet1/0/0 Huawei (link 1). Hal ini penting diperhatikan karena kalau salah pasang interface, link tidak akan bisa connect walaupun IP-nya sudah benar. Setelah semua IP di link point-to-point (Cisco–Huawei link 1 dan 2, Huawei–MikroTik, Cisco–MikroTik) dipasang sesuai tabel, semua router berhasil saling ping ke tetangga langsungnya, artinya koneksi fisik dan IP sudah benar semua.

Di Praktikum 4, OSPF berhasil membentuk hubungan antara tiga vendor yang berbeda (Cisco, Huawei, MikroTik). Ini menunjukkan kalau OSPF memang protokol open standard yang bisa jalan lintas vendor, asal area dan network-nya sesuai. Cisco berhasil punya dua neighbor ke Huawei (lewat dua link utama) dan satu neighbor ke MikroTik. Dengan cost 10 di link utama dan cost 100 di link via MikroTik, OSPF otomatis memilih jalur Cisco–Huawei langsung sebagai jalur utama, karena cost-nya lebih kecil. Ini sesuai dengan cara kerja OSPF yang pakai algoritma Dijkstra untuk cari jalur dengan total cost paling kecil. Berhasilnya ping end-to-end dari semua PC VLAN ke LAN Huawei (dan sebaliknya) membuktikan kalau OSPF sudah berhasil menyebarkan info routing ke semua router.

Di Praktikum 5, hasil pengujian failover menunjukkan bahwa selama masih ada minimal satu link utama Cisco–Huawei yang hidup, traffic tetap lewat jalur itu karena cost-nya (10) masih lebih kecil dari jalur backup via MikroTik (100). Ini membuktikan OSPF bisa langsung adjust rute tanpa putus koneksi, selama masih ada jalur dengan cost lebih kecil dari backup. Saat kedua link utama dimatikan bareng, OSPF otomatis mengalihkan traffic lewat jalur backup via MikroTik (PC VLAN → Cisco → MikroTik → Huawei → LAN Huawei), yang membuktikan proses convergence OSPF berjalan dengan baik.

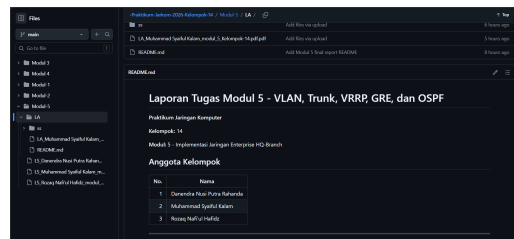
Setelah kedua link dihidupkan lagi, OSPF kembali update tabel routing dan balik milih jalur utama karena cost-nya lebih kecil. Ini menunjukkan OSPF memang dinamis, selalu menyesuaikan jalur terbaik sesuai kondisi topologi saat itu.

Secara keseluruhan, hasil yang didapat sudah sesuai dengan teori yang dipelajari, baik soal VLAN dan trunking, inter-VLAN routing dengan ROAS, DHCP server, maupun routing dinamis OSPF di lingkungan multivendor beserta mekanisme failover-nya. Hal yang perlu diperhatikan selama praktikum adalah ketelitian dalam memasang interface (sesuai disclaimer topologi), kesabaran nunggu proses booting perangkat, dan jangan lupa commit/save konfigurasi di setiap perangkat biar tidak hilang saat reload.

3 Hasil Tugas Modul

Hasil Tugas Modul Terdapat Pada Reeadme.md github Kelompok

<https://github.com/haikalramadhanasa/-Praktikum-Jarkom-2026-Kelompok-14/tree/main/Modul-5/LA>



Gambar 57: hasil Readme.md

4 Kesimpulan

Dari praktikum yang sudah dilakukan, bisa disimpulkan beberapa hal berikut:

1. Konfigurasi VLAN dan trunking di Cisco Switch berhasil, VLAN 10, 20, 30, dan 40 aktif dan bisa diteruskan dengan benar lewat trunk ke router.
2. Metode router-on-a-stick (ROAS) di Cisco Router berhasil membuat satu interface fisik bisa jadi gateway untuk beberapa VLAN sekaligus, dibuktikan dengan semua PC bisa ping ke gateway-nya dan ping antar-VLAN juga berhasil.
3. DHCP server di Cisco Router berhasil ngasih IP otomatis ke PC VLAN 10 dan 20 sesuai pool dan exclude address yang sudah diatur.
4. Semua link antarrouter (Cisco–Huawei link 1 dan 2, Huawei–MikroTik, Cisco–MikroTik) berhasil dikonfigurasi dan semua router bisa saling ping, walaupun ada perbedaan pemetaan interface dari topologi awal yang harus disesuaikan sesuai disclaimer.
5. OSPF berhasil dijalankan di tiga vendor berbeda (Cisco, Huawei, MikroTik), semuanya berhasil membentuk neighbor/peer dan saling sharing info routing, sehingga VLAN Cisco dan LAN Huawei bisa saling terhubung end-to-end.

6. Pengaturan OSPF cost berhasil membuat jalur utama (cost 10) lewat link langsung Cisco–Huawei dan jalur backup (cost 100) lewat MikroTik, sesuai dengan prinsip OSPF yang milih rute dengan cost paling kecil.
7. Pengujian failover membuktikan OSPF bisa konvergensi otomatis: traffic tetap lewat jalur utama selama salah satu link masih hidup, lalu pindah ke jalur backup via MikroTik kalau kedua link utama mati, dan balik lagi ke jalur utama begitu link tersebut dihidupkan kembali.